Kathmandu College of Technology

Lokanthali, Bhaktapur

Lab report on

Mathematics-I

As required for the partial fulfillment of

BCA 1st Semester

Affiliated to Tribhuvan university

Submitted by

Smriti Shrestha

Roll no: 25

Submitted to

Department of BCA

TABLE OF CONTENTS

LAB EXPERIMENT

NO.1

Function and Graph

Software used: Mathematica

Command used:

- Plot
- Inverse function
- Composite function

LAB EXPERIMENT
NO. 2

Matrix and Determinant

Software used: MATLAB

Commands used:

- Det (Determinant)
- Inv (Inverse)
- + (addition)
- - (subtraction)
- Transpose
- Rank
- Adjoint

LAB EXPERIMENT
NO.3

Scalar and Vector

Software used: MATLAB

Commands used:

- Norm(Magnitude)
- Cross (Cross Product)
- Dot (dot product)
- Scalar Triple product
- Vector triple product

LAB EXPERIMENT

NO. 4

Conic Section

Software used: MATHEMATICA

Commands used:

- Plot
- Show Graphics
- Contour plot

1. Construct any three 3*3 matrices X, Y and Z and perform the following operation using MATLAB.

a. det(A),det(B),det(C)
b. Inv(A),Inv(B),Inv(C)
c. A+B,B+C,C+A
d. A-B,B-C,C-A
e. Transpose(A)
f. Transpose(B)
g. Transpose(C)
h. X*Y,Y*Z and Z*X
i. Rank of X
j. Rank of Y
k. Rank of Z

Answer:

```
>> A=[1,2,3;4,5,6;7,8,9]

A =

     1     2     3
     4     5     6
     7     8     9

>> B=[5,1,9;-4, 5,-9;4,1,3]

B =

     5     1     9
    -4     5    -9
     4     1     3

>> C=[5,6,5;4,5,8;6,7,3]

C =

     5     6     5
     4     5     8
     6     7     3
```

```
>> det(A)

ans =

   6.6613e-16

>> det(B)

ans =

 -120.0000

>> det(C)

ans =

    1.0000

   >> inv(A)
   Warning: Matrix is close to singular or badly scaled. Results may be inaccurate. RCOND


   ans =

      1.0e+16 *

      -0.4504    0.9007   -0.4504
       0.9007   -1.8014    0.9007
      -0.4504    0.9007   -0.4504

   >> inv(B)

   ans =

      -0.2000   -0.0500    0.4500
       0.2000    0.1750   -0.0750
       0.2000    0.0083   -0.2417

   >> inv(C)

   ans =

     -41.0000   17.0000   23.0000
      36.0000  -15.0000  -20.0000
      -2.0000    1.0000    1.0000
```

```
>> A+B

ans =

     6     3    12
     0    10    -3
    11     9    12

>> B+C

ans =

    10     7    14
     0    10    -1
    10     8     6

>> C+A

ans =

     6     8     8
     8    10    14
    13    15    12
```

```
>> A-B

ans =

    -4     1    -6
     8     0    15
     3     7     6

>> B-C

ans =

     0    -5     4
    -8     0   -17
    -2    -6     0

>> C-A

ans =

     4     4     2
     0     0     2
    -1    -1    -6
```

```
>> transpose(A)

ans =

     1     4     7
     2     5     8
     3     6     9

>> transpose(B)

ans =

     5    -4     4
     1     5     1
     9    -9     3

>> transpose(C)

ans =

     5     4     6
     6     5     7
     5     8     3
```

```
>> rank(A)

ans =

     2

>> rank(B)

ans =

     3

>> rank(C)

ans =

     3
```

1. Construct any three space vectors a, b, c and obtain the following result using MATLAB.

- Norm of each vector
- All possible dot product
- Scalar triple product of given vector
- Vector triple product of given product.

```
>> a=[3,2,1]

a =

     3     2     1

>> b=[6,5,4]

b =

     6     5     4

>> c=[9,8,7
]

c =

     9     8     7


>> norm(a)

ans =

    3.7417

>> norm(b)

ans =

    8.7750

>> norm(c)

ans =

   13.9284
```

```
>> dot(a,b)

ans =

    32

>> dot(b,c)

ans =

   122

>> dot(c,a)

ans =

    50


>> cross(a,b)

ans =

     3    -6     3

>> cross(b,c)

ans =

     3    -6     3

>> cross(c,a)

ans =

    -6    12    -6

>> dot(a,cross(b,c))

ans =

     0

>> cross(a,cross(b,c))

ans =

    12    -6   -24
```

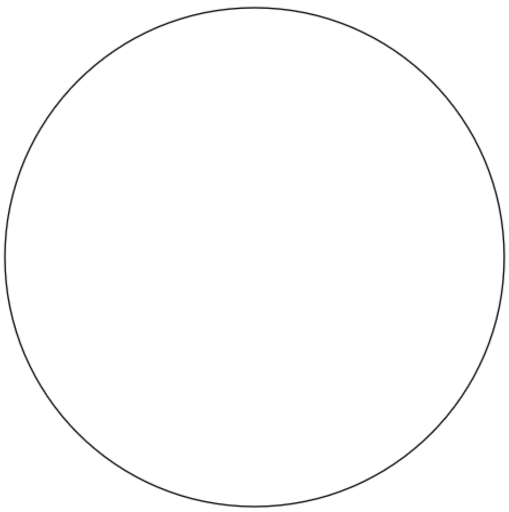
Plot circles having center (0,0) and radius

1. 3
2. 2

Answer:

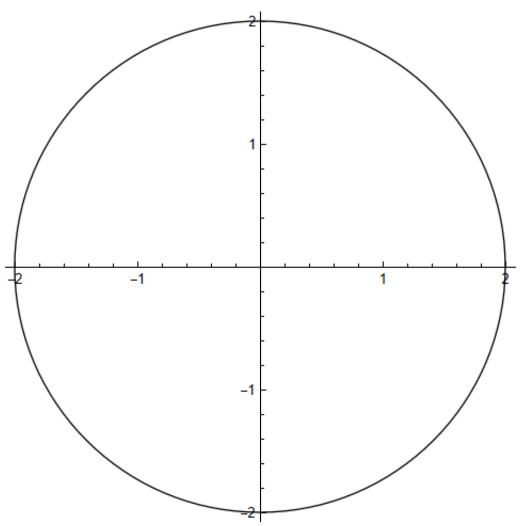
n[6]:= `Graphics[{Circle[{0, 0}, 3]}]`

ut[6]=



In[9]:= `Show[Graphics[{Circle[{0, 0}, 2]}], Axes → True, AxesStyle → Black]`

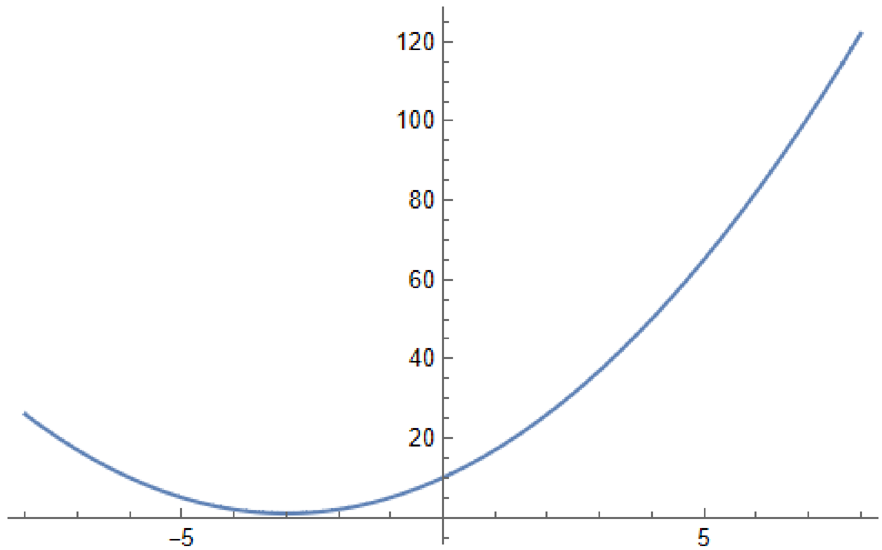
Out[9]=



Plot a parabola $x^2+6x+10$.
Answer:

```
In[11]:= Plot[Evaluate[x^2 + 6 x + 10], {x, -8, 8}]
```

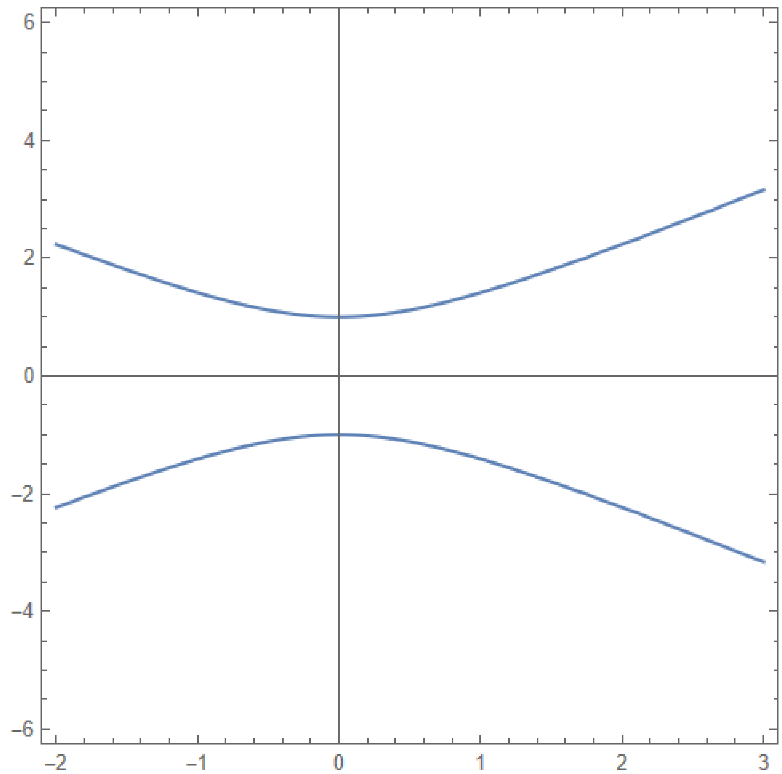
Out[11]=



Plot a hyperbola $y^2-x^2=1$
Answer:

```
In[15]:= ContourPlot[y^2 - x^2 == 1, {x, -2, 3}, {y, -6, 6}, Axes → True]
```

Out[15]=



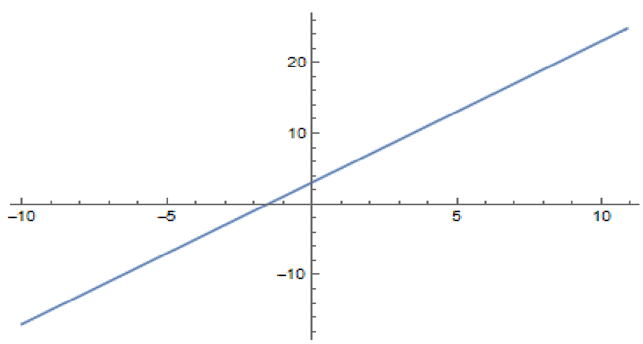
Let f: R→R defined by f(x)=2x+3, then find f(5).

```
In[5]:= F[x_] = 2 x + 3
        F[5]

Out[5]= 3 + 2 x

Out[6]= 13
```

Let f: R→R define f(x)=2x+3 and g: R→R define by g(x)=x^2 ,then find fog(x) and gof(x) and plot it in a graph.

```
In[18]:= F[x_] = 2 x + 3
         G[x_] = x^2
         F[G[x]]

Out[18]= 3 + 2 x

Out[19]= x^2

Out[20]= 3 + 2 x^2

In[21]:= G[F[x]]

Out[21]= (3 + 2 x)^2

In[26]:= Plot[3 + 2 x, {x, -10, 10.88}]
```

Out[26]=



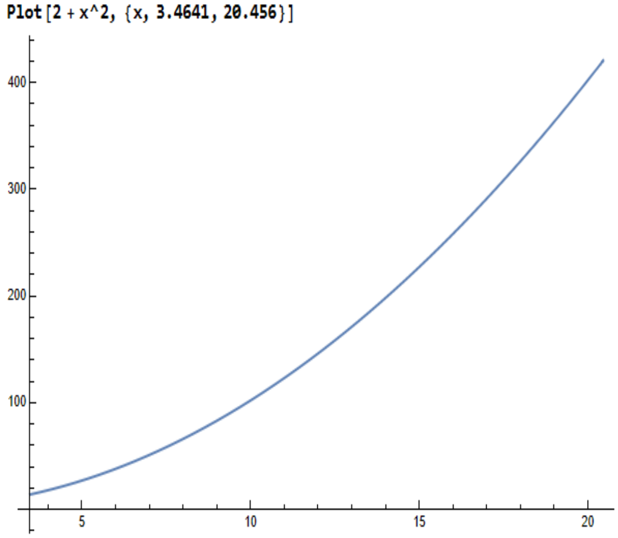
Let f:A→B define by $f(x)=x^2+2$, where A=(1,2,3), then find range.

```
In[27]:= F[x_] = x^2+2
         A= {1, 2, 3}
         F[A]

Out[27]= 2+x^2

Out[28]= {1, 2, 3}

Out[29]= {3, 6, 11}
```

If $f(x)= x+5$ and $g(x)=x^2-3$ then find (fog)(x)

```
In[1]:= f[x_] := x + 5
        g[x_] := x^2 - 3
        f[g[x]]
Out[3]= 2 + x^2

In[6]:= Plot[2 + x^2, {x, 3.4641, 20.456}]
```

Out[6]=



If f(x)= x+5 and g(x)=x^2-3 then find (gof)(x)

```
In[40]:= f[x_] := x + 5
         g[x_] := x^2 - 3
         g[f[x]]
Out[42]=
         -3 + (5 + x)^2

In[43]:= Plot[-3 + (5 + x)^2, {x, -80.56, 85.28}]
Out[43]=
```